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Thermodynamic modeling of the thulium–germanium binary system

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Abstract The thulium–germanium binary system has been critically assessed by means of the CALculation of PHAse Diagram technique through Thermo-Calc software package. The thulium–germanium binary system contains nine phases based on eight compounds, the intermediate phases with no solubility ranges: Tm_5Ge_3, Tm_5Ge_4, $Tm_{11}Ge_{10}$, TmGe, Tm_4Ge_5, $TmGe_{1.5}$, and $Tm_{0.9}Ge_2$ were treated as stoichiometric phases, while the compound with homogeneity range $TmGe_{1.8}$ ($TmGe_{1.8}$_LT and $TmGe_{1.8}$_HT) was modeled using two sublattices model as $(Ge,Tm)_{0.358}(Ge)_{0.642}$. The liquid phase was modeled as substitutional solution phase, in which the excess Gibbs free energy was formulated with Redlich–Kister polynomial. The germanium (diamond_A4) and thulium (hcp_A3) were kept as pure element phases since no solubility of germanium in thulium and vice versa. The calculations based on the thermodynamic modeling agree well with the phase diagram data and few experimental thermodynamic values from the literature.

Keywords Thermodynamic optimization · Phase diagram · CALPHAD method · Tm–Ge binary system · Redlich–Kister equation

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Introduction

The use of germanium technology is essentially linked to fiber optics communication network, infrared night vision systems, and semiconductor materials. These applications and else are attracting attention due to the high carrier mobilities of germanium [1–3] and its electrical properties. On the other side, the rare earth elements (REE) and their alloys and compounds with other elements are inexhaustible sources of new materials with unique chemical and physical properties. These properties make them indispensable in many fields of technology, such as magnetic and electrical materials, permanent magnet [4, 5], new energy technologies, catalysts, phosphors, solid-state lasers, and so on.

Germanides of rare earth elements (REE) and also alloys containing them represent a basis of materials promising for industrial applications and new technologies due to their unique electrophysical, chemical, and other valuable properties. These alloys are generally used in mirror coatings and Hall probes suitable for high-temperature applications, ohmic and rectifying contacts, infrared detectors, agents for refrigerators operating at near-room temperatures, etc. [6]. REE-Ge systems being studied more strongly due to their combination of the properties of the REE themselves and those of their alloys and compounds. Many intermetallics are formed in these systems. In this respect, the germanides R_5Ge_3 are the most refractory and congruently melting compounds in binary REE–Ge (REE = La–Lu) systems. They are all paramagnetic and show a typical variation in physicochemical properties above room temperature [7]. Concerning the pseudobinary R_5Ge_4 (R = rare earth elements) compounds type have received considerable attention due to their magnetic properties, e.g., giant magnetocaloric effect (MCE) [8, 9],

magnetostrictive effect [10], magnetoresistance effect [11, 12], and Griffiths-like phase behavior [13–16].

To satisfy the ever-increasing demands of new areas of technology for materials with a combination of electrophysical, magnetic, optical, thermal, and other properties, it is necessary to determine the phase diagram and thermodynamic properties of the chemical compounds and phases which are carriers of such properties.

The aim of the present paper is to evaluate the Tm–Ge system using the CALPHAD method and the Thermo-Calc software package based on available experimental data. The current work is a continuation of similar studies of REE-Ge (REE = Y [17], Ho [18], Yb [19] and Er [40]) already calculated in our laboratory and the first in thermodynamic modeling of Tm–Ge binary system. The process of modeling allows us to calculate the phase equilibria and thermodynamic properties of the liquid phase and intermetallic compounds such as the mixing enthalpies, the entropies of formation, and the enthalpies of formation for compounds that were not examined.

Evaluation of data from the literature

Phase diagram

The thulium–germanium phase equilibria have been constructed for the first time by Eremenko et al. [20] and redrawn by Massalski [21] (Fig. 1). Eremenko et al. investigated the Tm–Ge system by methods of microscopic, X-ray diffraction, and differential thermal analyses.

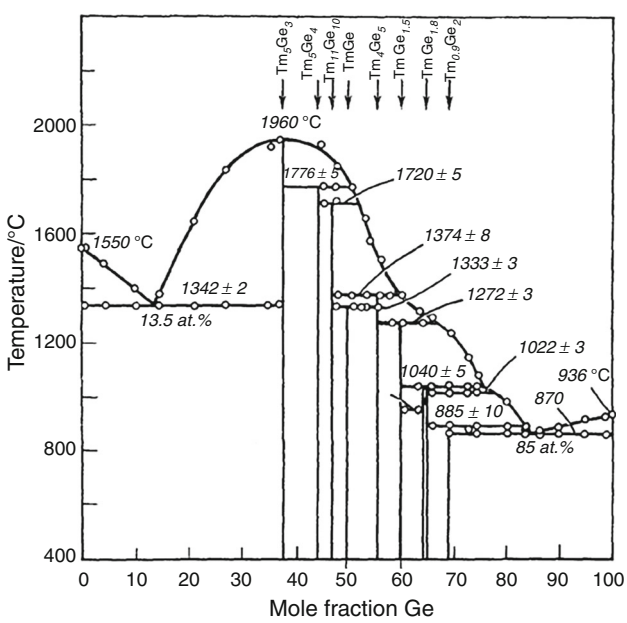

Fig. 1 Experimental phase diagram of the Tm–Ge binary system [20]

This phase diagram shows the existence of nine intermediate phases based on eight germanides which were: Tm_5Ge_3, Tm_5Ge_4, $Tm_{11}Ge_{10}$, TmGe, Tm_4Ge_5, $TmGe_{1.5}$, $TmGe_{1.8}$, and $Tm_{0.9}Ge_2$. The most thulium-rich germanide Tm_5Ge_3 melts congruently at 1960 °C. The monogermanide TmGe is formed by a peritectoid reaction at 1333 ± 3 °C. All of the remaining thulium germanides Tm_5Ge_4, $Tm_{11}Ge_{10}$, Tm_4Ge_5, $TmGe_{1.5}$, $TmGe_{1.8}$, and $Tm_{0.9}Ge_2$ melt incongruently by peritectic reactions at 1776 ± 5, 1720 ± 5, 1374 ± 8, 1272 ± 3, 1022 ± 3, and 885 ± 10 °C, respectively. Tm_5Ge_4 crystallizes together with the thulium-base solid solution at 1342 ± 2 °C, and the composition of the eutectic is 13.5 at.% Ge. The most germanium-rich compound together with the germanium solid solution crystallizes according to the eutectic reaction at 870 °C and 85 at.%. All of the thulium germanides are stoichiometric phases, except the compound with a composition of $TmGe_{1.8}$ which has a narrow area of homogeneity. This germanide is dimorphous and the temperatures of its polymorphic transformation are different (950 and 1022 °C), respectively.

Thermodynamic data

The only investigation of the standard enthalpies of formation in the thulium–germanium binary system was reported by Meschel and Kleppa [22]. The standard enthalpies of formation of Tm_5Ge_3 ($-$ 91.3 ± 1.9 kJ $mol^{-1}at^{-1}$) have been reported using high-temperature direct synthesis calorimetry at 1373 ± 2 K.

Crystallography

A basic characteristic of the solid phases is the crystal structure, which although it does not have a direct representation in the phase diagrams significantly influences their topology because it is related to the thermodynamic stability. The crystallographic data for the intermediate phases formed in the thulium–germanium binary system are reported in Table 1.

Assessment procedure

The application of CALPHAD method using the Thermo-Calc software developed by Jansson [23] and Sundman et al. [24], in the optimization of the model parameters of the Gibbs free energy expressions of each phase in the system can be divided to two substeps. The first is the choice of thermodynamic models for the Gibbs energy of the individual phases. The second is adjusting the parameters *a* and *b* (mentioned in thermodynamic models section) related to the phases in order to describe the liquidus and invariant reactions. The modeling procedure was carried out as

Table 1 Crystallographic data of the phases in Tm–Ge binary system

Compound	at.% Ge	Crystal system	Structure type	Space group	References
Tm_5Ge_3	37.5	Hexagonal	Mn_5Si_3	*P6₃/mcm*	[32, 33]
Tm_5Ge_4	44.4	Orthorhombic	Sm_5Ge_4	*Pnma*	[20, 34, 38]
$Tm_{11}Ge_{10}$	47.6	Tetragonal	$Ho_{11}Ge_{10}$	–	[20, 35]
TmGe	50.0	Orthorhombic	CrB	–	[20]
Tm_4Ge_5	55.6	–	–	–	–
$TmGe_{1.5}$	60.0	Hexagonal	AlB_2	–	[20, 37]
$TmGe_{1.8}$_LT	64.2	–	$ErGe_{1.83}$	*Cmcm*	[39]
$TmGe_{1.8}$_HT	64.2	–	–	–	–
$Tm_{0.9}Ge_2$	68.9	Orthorhombic	*ZrSi2*	–	[36]

follows: The first stage of the optimization began with introducing a file of the stability of the liquid phase. The liquid phase was treated as a substitutional solution using Redlich–Kister polynomial [25]. In the case of the compounds, the congruent intermetallic compound Tm_5Ge_3 was taken into account first. Then the other intermetallic phases were introduced in the modeling one by one. Next, each piece of experimental data related to invariant equilibria is added to the calculation with certain weight. The weights were changed systematically during the assessment until all utilized experimental data were accounted for within the experimental uncertainties claimed in the literature or evaluated in this work. Finally, all parameters are adjusted again and assessed simultaneously in the final calculation steps until most of experimental data were accounted for within the expected uncertainty limits. The optimization of the parameters is completed as soon as the square sum of the differences between measured and computed values is reduced to the minimum.

No excess term was optimized for the solid solutions because no solubility of Tm in Ge or Ge in Tm was experimentally determined. For the thulium–germanium system, the optimization was difficult due to the similar temperatures of some invariant reactions.

In this work, the optimization was carried out in two steps. In the first treatment, $TmGe_{1.8}$ was assumed to be a stoichiometric compound. And in the second, it was treated using a two-sublattice model [26, 27]. The parameters obtained from the first treatment were used as starting values for the second.

Thermodynamic models

Pure elements

The Gibbs free energy function for the pure element i (i = Tm, Ge) in its stable state of the phase Φ (Φ = hcp_A3 and Diamond_A4) is described by an equation of the following form:

$$G^{\Phi} - H^{\Phi}_{\mathrm{SER}} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where $H_i^{\mathrm{SER}}(298.15\ \mathrm{K})$ is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state, hcp_A3 for thulium and diamond_A4 for germanium, and T is the absolute temperature; a, b, c,…, h are the coefficients. The Gibbs free energy of the element i, in its SER state, is denoted by *GHSERi* as follows:

$$\mathrm{GHSER_{Ge}} = {}^0G^{\mathrm{Diamond_A4}}_{\mathrm{Ge}} = G^{\mathrm{Diamnd_A4}}_{\mathrm{Ge}} - H^{\mathrm{Diamnd_A4}}_{\mathrm{Ge}}(298.15\ \mathrm{K}) \quad (2)$$

$$\mathrm{GHSER_{Tm}} = {}^0G^{\mathrm{hcp_A3}}_{\mathrm{Tm}} = G^{\mathrm{hcp_A3}}_{\mathrm{Tm}} - H^{\mathrm{hcp_A3}}_{\mathrm{Tm}}\ (298.15\ \mathrm{K}) \quad (3)$$

In the present optimization, the Gibbs free energies functions are taken from the Scientific Group Thermo data Europe (SGTE) database compiled by Dinsdale [28].

Solution phases

The molar Gibbs free energy of the liquid phase can be expressed as the sum of the reference part, the ideal part and the excess part:

$$G^{\mathrm{liquid}}_{\mathrm{m}} = {}^{\mathrm{ref}}G^{\mathrm{liquid}} + {}^{\mathrm{id}}G^{\mathrm{liquid}} + {}^{\mathrm{ex}}G^{\mathrm{liquid}} \quad (4)$$

$$G^{\mathrm{liquid}}_{\mathrm{m}} = x_{\mathrm{Tm}}\,{}^0G^{\mathrm{liquid}}_{\mathrm{Tm}} + x_{\mathrm{Ge}}\,{}^0G^{\mathrm{liquid}}_{\mathrm{Ge}} + \mathrm{RT}(x_{\mathrm{Tm}}\ln x_{\mathrm{Tm}} + x_{\mathrm{Ge}}\ln x_{\mathrm{Ge}}) + {}^{\mathrm{ex}}G^{\mathrm{liquid}}_{\mathrm{m}} \quad (5)$$

where R is the ideal gas constant, T is the temperature in Kelvin, x_{Tm} and x_{Ge} are the mole fraction of the elements Tm and Ge, respectively.

The excess Gibbs energy, ${}^{\mathrm{ex}}G^{\mathrm{liquid}}_{\mathrm{m}}$ is described by the Redlich–Kister polynomial [22] as follows:

$${}^{\mathrm{ex}}G^{\mathrm{liquid}}_{\mathrm{m}} = x_{\mathrm{Tm}}x_{\mathrm{Ge}}\sum_{i=0}^{n}{}^iL^{\mathrm{liquid}}_{\mathrm{Tm,Ge}}(x_{\mathrm{Tm}} - x_{\mathrm{Ge}})^i \quad (6)$$

The ith interaction parameters between components Tm and Ge, ${}^iL^{\mathrm{liquid}}_{\mathrm{Tm,Ge}}$ were temperature dependent are taken as a linear function of temperature: Redlich–Kister equation [22] as follows:

$${}^iL^{\mathrm{liquid}}_{\mathrm{Tm,Ge}} = a_{\mathrm{i}} + b_{\mathrm{i}}T \quad (i = 0, 1, 2, \ldots) \quad (7)$$

with the excess entropy $s = -b_{\mathrm{i}}$.

No excess term of the Gibbs free energy was optimized for terminal solid solutions, because no solubility of Ge in Tm and Tm in Ge was reported.

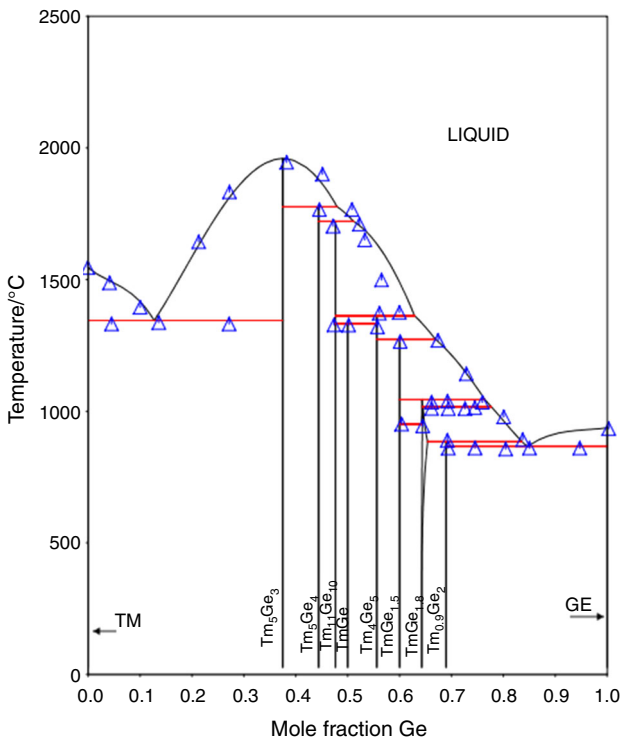

Fig. 2 Calculated Tm–Ge binary phase diagram with the present description compared with the experimental data [20]

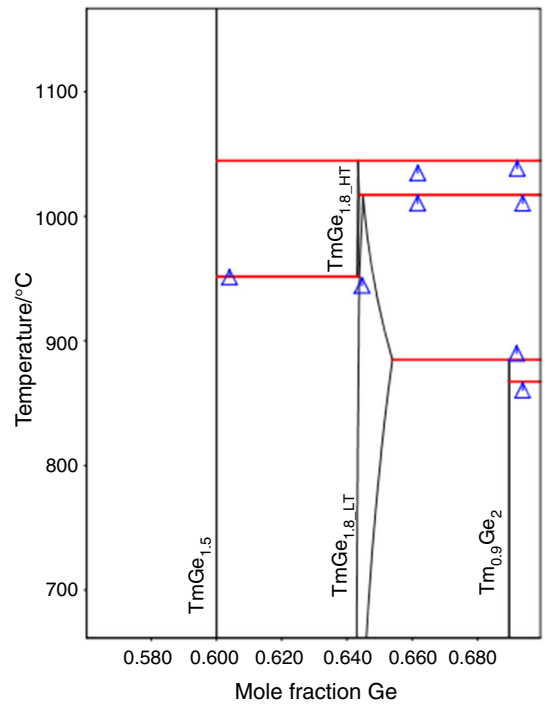

Fig. 3 Enlarged phase diagram of the $TmGe_{1.8}$ phase with experimental data [20]

Stoichiometric compounds

In the Tm–Ge binary system seven intermediate compounds: Tm_5Ge_3, Tm_5Ge_4, $Tm_{11}Ge_{10}$, TmGe, Tm_4Ge_5, $TmGe_{1.5}$, and $Tm_{0.9}Ge_2$ are treated as stoichiometric phases and described as A_pB_q.

The Gibbs free energy of each phase is denoted as ${}^0G_{Tm_pGe_q}$ and described by the following formula:

$$ {}^0G_{Tm_pGe_q} = \frac{p}{p+q}\,{}^0G_{Tm} + \frac{q}{p+q}\,{}^0G_{Ge} + a + bT \tag{8} $$

where ${}^0G_{Tm}$ and ${}^0G_{Ge}$ are the Gibbs free energies of the pure elements thulium and germanium, respectively; a and b are optimized for each of the compounds in the present work. The coefficients a can be estimated according to the corresponding experimental enthalpies of formation, then evaluated by taking into account both phase equilibria and thermodynamic data.

Compounds with homogeneity range

The $TmGe_{1.8}$_LT and $TmGe_{1.8}$_HT phases are a non-stoichiometric compounds which have a homogeneity range on the Ge-rich side of the strict stoichiometry $TmGe_{1.8}$, was treated as the formula $(Ge,Tm)_{0.358}(Ge)_{0.642}$ by a two-sublattice model [26, 27]. The Gibbs free energy function per mole (m) of the formula unit is given by the following expression:

$$ \begin{aligned} {}^0G_m^{TmGe1.8} &= G_m^{TmGe1.8} - H^{SER}(298.15\ K) \\ &= y_{Ge}\,{}^0G_{Ge:Ge}^{TmGe1.8} + y_{Tm}\,{}^0G_{Tm:Ge}^{TmGe1.8} \\ &\quad + RT(y_{Tm}\ln y_{Tm} + y_{Ge}\ln y_{Ge}) \\ &\quad + y_{Tm}y_{Ge}\sum_i {}^{\upsilon}L_{Ge,Tm:Ge}(y_{Ge} - y_{Tm})^{\upsilon} \end{aligned} \tag{9} $$

where y_{Tm} and y_{Ge} are the site fraction of component Tm and Ge located on two sublattices, and the parameter ${}^0G_{i:j}$ represents the Gibbs free energy of the compound when the two sublattices are occupied by element i or j (i, j = Tm, Ge). ${}^{\upsilon}L_{i:j}$ are the interaction parameters between Tm and Ge in the second or first sublattice, when the other sublattice is occupied by element i or j.

Results and discussion

The computed Tm–Ge phase diagram using the present set of thermodynamic parameters is compared with the literature data reported by Eremenko et al. [20]. In Fig. 2. The detailed phase relationships from 56 to 70 at.% Ge and temperatures from 700 to 1100 °C are further shown together with experimental data in Fig. 3. The phase diagram obtained in this work is in good agreement with the established one by Eremenko et al. [20].

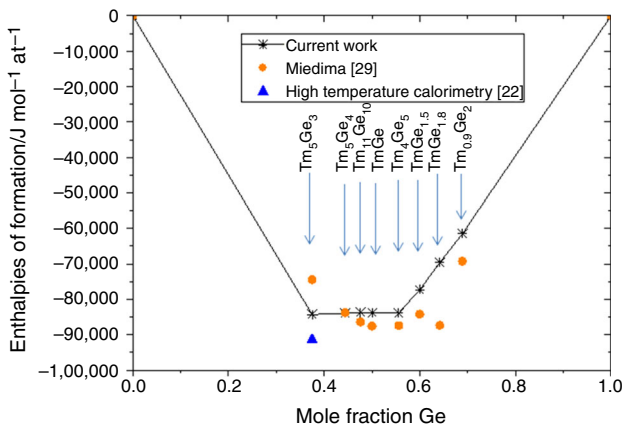

Fig. 4 Comparison between the computed enthalpies of formation, the predicted ones [29], and the experimental data of the compound Tm_5Ge_3 [22]

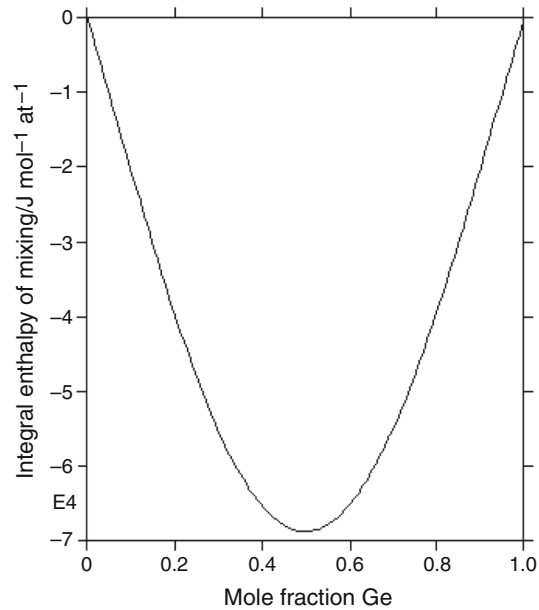

Fig. 5 Calculated enthalpies of mixing of liquid phase at 1973 K

Except the Tm_5Ge_3 compound, we did not find in the literature any enthalpy of formation relative to the remaining compounds. We give in this work, for the first time, the calculated values of their enthalpies of formation. Figure 4 and Table 2 present the calculated enthalpies of formation of Tm–Ge intermediate phases from CALPHAD modeling together with the predicted values by the Miedema model [29] and those from the experimental data determined by Meschel and Kleppa [22]. A reasonable agreement is noted between the calculated results and the experimental values determined by calorimetry for Tm_5Ge_3 phase. The calculated enthalpies of formation for many compounds are consistent with the predicted ones by Miedema model [29].

The computed enthalpy of mixing of liquid phase calculated in this work over the whole range of composition at 1973 K is presented in Fig. 5; owing to the lack of

Table 2 Comparison between the enthalpies of formation of the intermetallic compounds in the Tm–Ge system derived from the present optimization and taken from the literature

Intermetallic compounds	Enthalpy of formation/kJ mol^{-1} at^{-1}	Technique used	Entropy of formation/J mol^{-1} at^{-1} K^{-1}	References
Tm_5Ge_3	− 84.32	Optimization	− 2.14	Current work
	− 91.30 ± 1.9	Experimental		[22]
	− 74.42	Prediction		[29]
Tm_5Ge_4	− 83.94	Optimization	− 0.72	Current work
	− 83.71	Prediction		[29]
$Tm_{11}Ge_{10}$	− 83.70	Optimization	− 0.42	Current work
	− 86.37	Prediction		[29]
TmGe	− 83.80	Optimization	− 1.5	Current work
	− 87.57	Prediction		[29]
Tm_4Ge_5	− 83.84	Optimization	− 4.04	Current work
	− 87.44	Prediction		[29]
$TmGe_{1.5}$	− 77.30	Optimization	− 2.67	Current work
	− 84.18	Prediction		[29]
$TmGe_{1.8}$_HT	− 69.48	Optimization	− 1.83	Current work
	− 87.37	Prediction		[29]
$TmGe_{1.8}$_LT	− 69.60	Optimization	− 1.93	Current work
	− 87.37	Prediction		[29]
$Tm_{0.9}Ge_2$	− 61.46	Optimization	− 2.36	Current work
	− 69.24	Prediction		[29]

Table 3 Optimized thermodynamic parameters of the Tm–Ge system using Redlich–Kister equation [25]

Phases	Thermodynamic model	Parameters
Liquid	(Tm, Ge)	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Tm,Ge}} = -275,450.516 + 19.8279T$
		${}^{1}L^{\mathrm{Liq}}_{\mathrm{Tm,Ge}} = 2031.9279$
		${}^{2}L^{\mathrm{Liq}}_{\mathrm{Tm,Ge}} = 79,130.1539$
hcp_A3	$(\mathrm{Tm, Ge})_1(V_a)^{*}{}_1$	No excess term optimized
Diamond_A4	(Ge)	No excess term optimized
Tm_5Ge_3	$Tm_{0.625}$:$Ge_{0.375}$	$G^{\mathrm{Tm5Ge3}}_{\mathrm{Tm,Ge}} - 0.625 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.375 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.625 * \mathrm{GHSERTm} + 0.375 * \mathrm{GHSERGe} - 84,328.2714 + 2.1448 * T$
Tm_5Ge_4	$Tm_{0.556}$:$Ge_{0.444}$	$G^{\mathrm{Tm5Ge4}}_{\mathrm{Tm,Ge}} - 0.556 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.444 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.556 * \mathrm{GHSERTm} + 0.444 * \mathrm{GHSERGe} - 83,946.9470 + 0.7257 * T$
$Tm_{11}Ge_{10}$	$Tm_{0.523}$:$Ge_{0.476}$	$G^{\mathrm{Tm11Ge10}}_{\mathrm{Tm,Ge}} - 0.523 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.476 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.523 * \mathrm{GHSERTm} + 0.476 * \mathrm{GHSERGe} - 83,707.8089 + 0.4280 * T$
TmGe	$Tm_{0.5}$:$Ge_{0.5}$	$G^{\mathrm{TmGe}}_{\mathrm{Tm,Ge}} - 0.5 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.5 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.5 * \mathrm{GHSERTm} + 0.5 * \mathrm{GHSERGe} - 83,807.6603 + 1.5001 * T$
Tm_4Ge_5	$Tm_{0.444}$:$Ge_{0.556}$	$G^{\mathrm{Tm4Ge5}}_{\mathrm{Tm,Ge}} - 0.444 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.556 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.444 * \mathrm{GHSERTm} + 0.556 * \mathrm{GHSERGe} - 83,842.8254 + 4.0494 * T$
$TmGe_{1.5}$	$Tm_{0.4}$:$Ge_{0.6}$	$G^{\mathrm{TmGe1.5}}_{\mathrm{Tm,Ge}} - 0.4 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.6 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.4 * \mathrm{GHSERTm} + 0.6 * \mathrm{GHSERGe} - 77,304.6102 + 2.6772 * T$
$TmGe_{1.8}$_HT	$(Ge,Tm)_{0.357}$:$(Ge)_{0.642}$	$G^{\mathrm{TmGe1.8_HT}}_{Tm,Ge} - 0.357 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.642 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.357 * \mathrm{GHSERTm} + 0.642 * \mathrm{GHSERGe} - 69,482.5413 + 1.8381 * T$
		$G^{\mathrm{TmGe1.8_HT}}_{\mathrm{Ge,}Ge} - H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 26,393.3131$ + GHSERGE - 497.6504
		$L^{\mathrm{TmGe1.8_HT}}_{\mathrm{Ge,Tm:Ge}} = -13,091.5540$
$TmGe_{1.8}$_LT	$(Ge,Tm)_{0.357}$:$(Ge)_{0.642}$	$G^{\mathrm{TmGe1.8_LT}}_{\mathrm{Tm,Ge}} - 0.357 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.642 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.357 * \mathrm{GHSERTm} + 0.642 * \mathrm{GHSERGe} - 69,600.0503 + 1.9393 * T$
		$G^{\mathrm{TmGe1.8_LT}}_{\mathrm{Ge,Ge}} - H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 26,393.3131$ + GHSERGE - 537.7988
		$L^{\mathrm{TmGe1.8_LT}}_{\mathrm{Ge,Tm:Ge}} = -15,956.4431$
$Tm_{0.9}Ge_2$	$Tm_{0.310}$:$Ge_{0.689}$	$G^{\mathrm{Tm0.9Ge2}}_{\mathrm{Tm,Ge}} - 0.310 * {}^{298.15}H^{\mathrm{HCP_A3}}_{\mathrm{Tm}} - 0.689 * {}^{298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}}$
		$= 0.310 * \mathrm{GHSERTm} + 0.689 * \mathrm{GHSERGe} - 61,467.5982 + 2.3605 * T$

* (V_a) for vacancy

experimental determinations, no comparison with the literature is possible.

The optimized thermodynamic parameters for all the phases in the Tm–Ge binary system obtained in the present work are summarized in Table 3.

The calculated compositions and temperatures of the invariant reactions in this system are compared with the experimental ones in Table 4. It can be seen that Our calculated values concerning the six peritectic reactions, two eutectic reactions, and a peritectoid reaction which occur in this system are in excellent agreement with the experimental temperatures and compositions taken into account the errors given by Eremenko et al. [20] in the temperatures of some invariant reactions. The difference between the calculated and the measured invariant equilibrium temperatures is less than 5 K. Concerning the $TmGe_{1.8}$ phase which melts peritectically at 1040 K with two allotropic forms ($\alpha TmGe_{1.8}$ and $\beta TmGe_{1.8}$), the temperatures of its polymorphic transformation (950 and 1022 °C) are in good agreement with the literature data [20].

In Fig. 6, the Gibbs energy of mixing of the liquid phase was calculated at different temperatures: 2000–7000 K with Redlich–Kister model. This figure shows that the Gibbs energies of mixing for the liquid phase become more negative when the temperature increases. In Fig. 7 we plot the phase diagram when the liquid phase is suspended. We note that when the liquid phase is not considered during the calculation of the phase diagram at high temperature, the

Table 4 Comparison between the experimental invariant reactions [20] and the calculated ones in the Tm–Ge system

Reactions	Reaction type	Literature [17]		This work	
		T/°C	*x* liq/at.% Ge	*T*/°C	*x* liq/at.% Ge
L ↔ Tm	Melting	1550	0.00	1550	0.00
L ↔ Tm + Tm_5Ge_3	Eutectic	1342 ± 2	0.135	1345	0.127
L ↔ Tm_5Ge_3	Congruent	1960	0.375	1960	0.375
Tm_5Ge_3 ↔ L + Tm_5Ge_4	Peritectic	1776 ± 5	0.500	1777	0.479
Tm_5Ge_4 ↔ L + $Tm_{11}Ge_{10}$	Peritectic	1720 ± 5	0.520	1722	0.511
$Tm_{11}Ge_{10}$ + L ↔ Tm_4Ge_5	Peritectic	1374 ± 8	0.600	1363	0.629
TmGe ↔ $Tm_{11}Ge_{10}$ + Tm_4Ge_5	Peritectoid	1333 ± 3	0.500	1333	0.500
Tm_4Ge_5 + L ↔ $TmGe_{1.5}$	Peritectic	1272 ± 3	0.670	1274	0.668
$TmGe_{1.8}$_HT ↔ $TmGe_{1.8}$_LT + $TmGe_{1.5}$	Peritectoid	950	0.642	952	0.642
$TmGe_{1.8}$_HT ↔ $TmGe_{1.8}$_LT + L	Metatectic	1022 ± 3	0.643	1017	0.643
$TmGe_{1.5}$ + L ↔ $TmGe_{1.8}$_HT	Peritectic	1040 ± 5	0.750	1045	0.764
Tm_5Ge_9_LT + L ↔ $Tm_{0.9}Ge_2$	Peritectic	885 ± 10	0.830	885	0.835
L ↔ $Tm_{0.9}Ge_2$ + Ge	Eutectic	870	0.850	867	0.846
L ↔ Ge	Melting	936	1.00	936	1.00

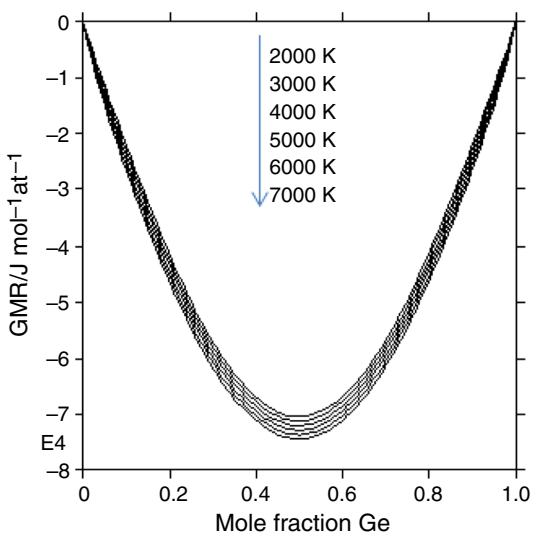

Fig. 6 Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (2000–7000 K) using the Redlich–Kister model [25]

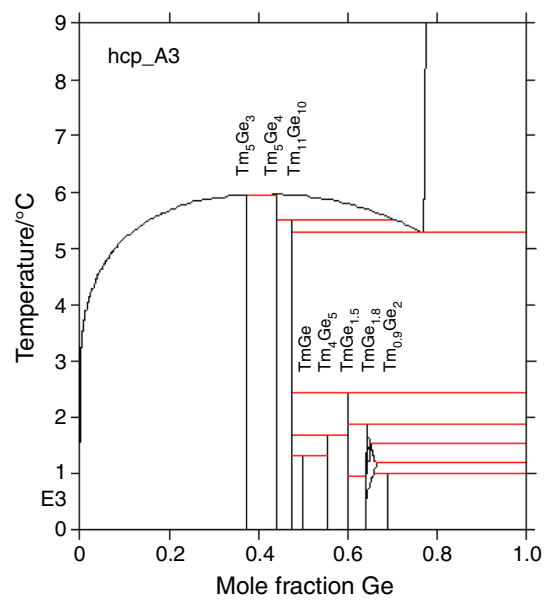

Fig. 7 Calculated Tm–Ge phase diagram when the liquid phase is suspended

stoichiometric phases disappear and only the hcp_A3 solid solution is calculated on the whole germanium composition range. As indicated in Refs. [30, 31], our optimized parameters are reliable.

Finally, we mention that during the optimization, no formation of an inverted miscibility gap was detected at very high temperatures.

Conclusions

Based on the experimental measurements from studies, a set of self-consistent thermodynamic parameters for the description of the Tm–Ge system was derived by using CALPHAD technique in which the Gibbs energy functions were optimized. The experimental data of the phase equilibria and the thermodynamic properties in the system were

critically evaluated. The computed results reproduce all the phase relations in this system and agree well with the available experimental data. For the studied system, accurate thermodynamic data are rather scarce or lacking, in particular the enthalpies of formation and the mixing enthalpies of the liquid phase. In this work, the obtained thermodynamic parameters allow us to reproduce the phase diagram of the Tm–Ge system well and to determine the entropies and enthalpies of formation relative to the intermetallic compounds for the first time.

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